

DATABASE DESIGN & NORMALIZATION

SCENARIO 1: SMART CAMPUS HOSTEL MANAGEMENT

Entities and Attributes

Entity	Attributes
Hostel	PK: hostel_id, hostel_name
Room	room_id [PK], hostel_id [FK], capacity
Student	student_id [PK], name, room_id [FK]
Student Phone	phone_id [PK], student_id [FK], phone
Warden	warden_id [PK], name
Hostel_warden	hostel_id [FK], warden_id [FK]
Maintenance_Staff	staff_id [PK], name
Complaint	complaint_id [PK], student_id [FK], room_id [FK], description, status, created_at, updated_at
Complaint_Assignment	complaint_id [FK], staff_id [FK]

Relationships

- Hostel → Room = 1:N
- Room → Student = 1:N
- Student → Student_Phone = 1:1
- Hostel ↔ Warden = M:N
- Student → Complaint = 1:N
- Complaint ↔ Maintenance_Staff = M:N

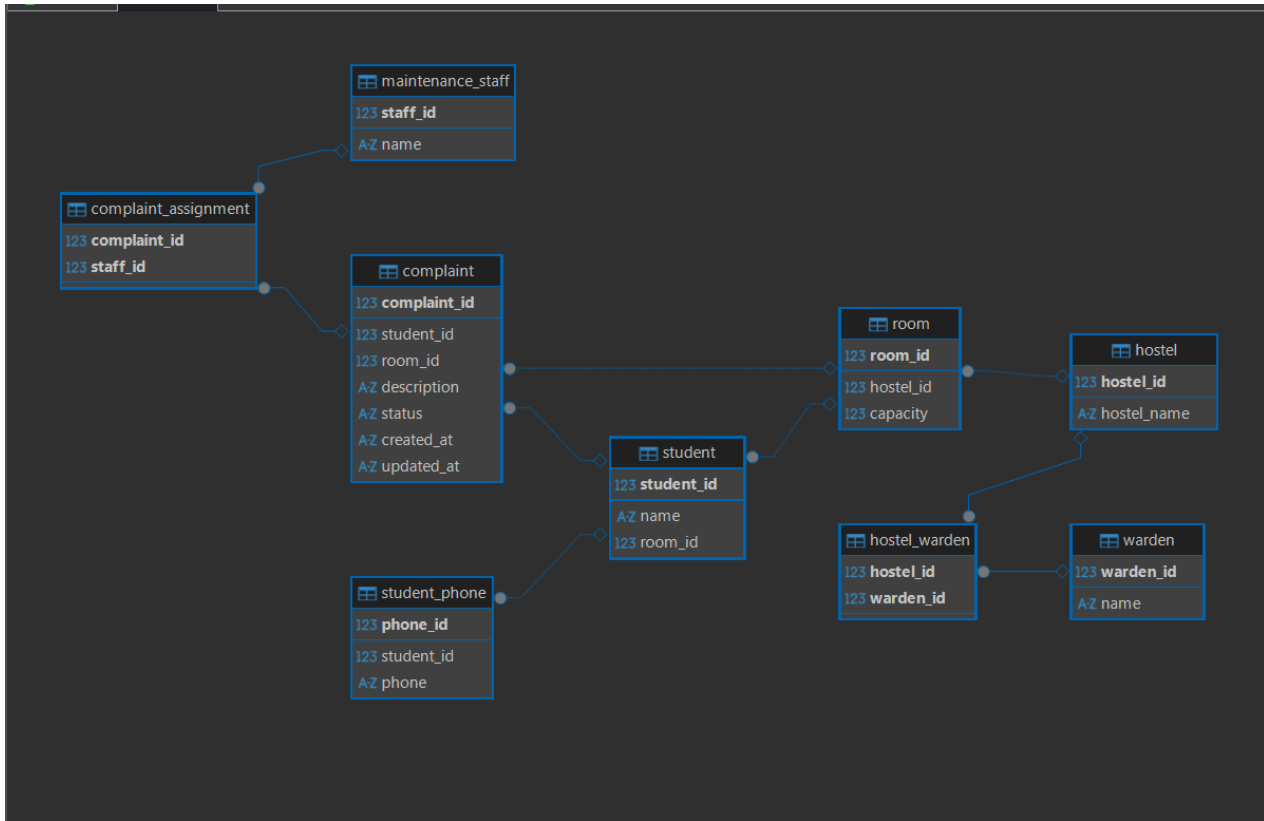
Normalization

1NF: All attributes are atomic. Repeating groups are removed.

2NF: No partial dependency; non-key attributes fully depend on the primary key.

3NF: No transitive dependency; redundancy is minimized by storing data in appropriate tables

ER Diagram



SCENARIO 2: MINI E-COMMERCE PLATFORM

Entities and Attributes

Entity	Attributes
Customers	customer_id [PK], name, email, phone
Category	category_id (PK), category_name
Product	product_id [PK], name, category_id [FK], price
Orders	order_id [PK], customer_id [FK], order_date, status
Order_Items	order_id [PK], product_id [PK], quantity
Payment	payment_id [PK], order_id [FK], amount, payment_method, payment_date

Relationships

- Category → Product = 1:N
- Customer → Orders = 1:N
- Orders ↔ Product = M:N
- Orders → Payment = 1:N

ER diagram

